DNSC 6330 · Responsible Machine Learning · George Washington University

Group 2

A Responsible AI Audit · 2024 HMDA Mortgage Data

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Q1: Optimization Objective

Optimization Objective: ROC-AUC with Threshold = 0.7

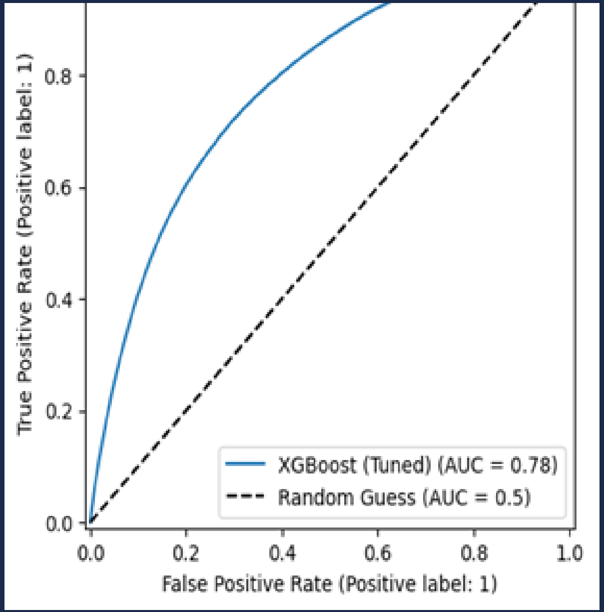
Objective

- Maximize **ROC-AUC** to ensure strong **ranking performance** between approved (1) and denied (0) mortgage applications
- Use **decision threshold = 0.7** for final classification

Key Insight

- Significant improvement in **ranking ability (AUC)**
- Smaller gain in **threshold-based performance (F1)**

→ Indicates **non-linear relationships captured by XGBoost**



Why AUC + Why Threshold = 0.7?

Why ROC-AUC?

- Measures model performance **across all thresholds**, not just one
- Captures **global ranking quality**, critical in lending decisions
- More robust than accuracy under **class imbalance**
- Aligns with regulatory expectations:
  - Evaluates whether the model **systematically separates groups** before thresholding
- From results: **+0.1163 AUC gain** shows strong improvement in predictive ordering

Why Threshold = 0.7?

- Conservative cutoff → reduces **false approvals (risk exposure)**
- Aligns with **risk-sensitive lending policy**
- Moves decision boundary to favor **precision over recall**
- Reflects business need:
  - Cost of approving a bad loan > denying a good loan

Trade-off

- Higher threshold →
  - ↓ False positives (good for lender risk)
  - ↑ False negatives (potential fairness concern)

Q2: Known Failure Modes

Where does the model break down — and who is harmed when it does?

◈ SHAP

Global & Local Importance

◉ SHAP

Waterfall Explanations

◆ DiCE

Counterfactual Analysis

◇ PSI

Feature Drift Testing

- Failure Mode 1 → Proxy Bias
- Failure Mode 2 → Risk Override
- Failure Mode 3 → Recourse Failure
- Failure Mode 4 → Feature Drift

MODE 01

Proxy Discrimination via Neighborhood Income

WHAT HAPPENS

The model uses ffiec_msa_md_median_family_income, which is the median income of the applicant's census tract, as an input feature. Because of residential segregation, this penalizes applicants for where they live rather than their personal financial profile.

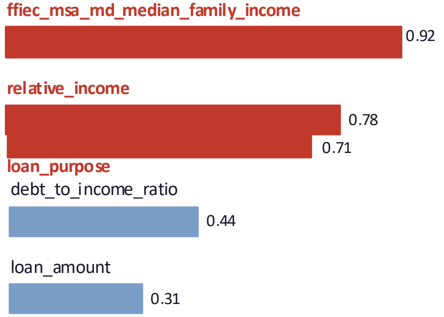
WHO IS HARMED

African-American applicants, who are disproportionately and systemically located in lower-income census tracts due to historical redlining. Race never enters the model explicitly but it enters through the zip code.

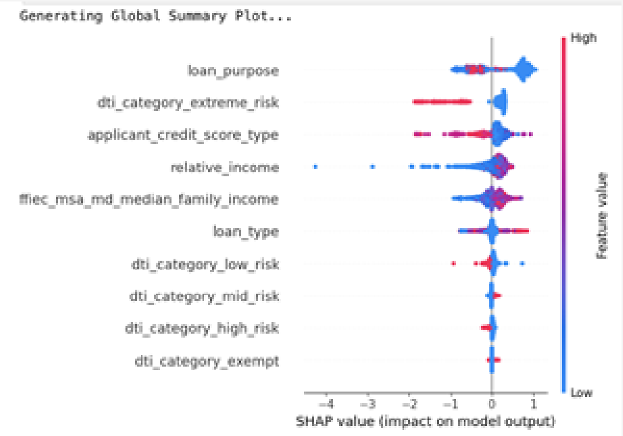
SEVERITY: HIGH Potential ECOA violation

Evidence from Our SHAP Analysis

Top Features by Mean |SHAP| Value



▲ Proxy features flagged in red — SHAP shows these drive predictions across most applicants



MODE 02

SEVERITY: HIGH

Risk Override · DTI Bypass

SHAP · Local (Waterfall)

WHAT FAILS

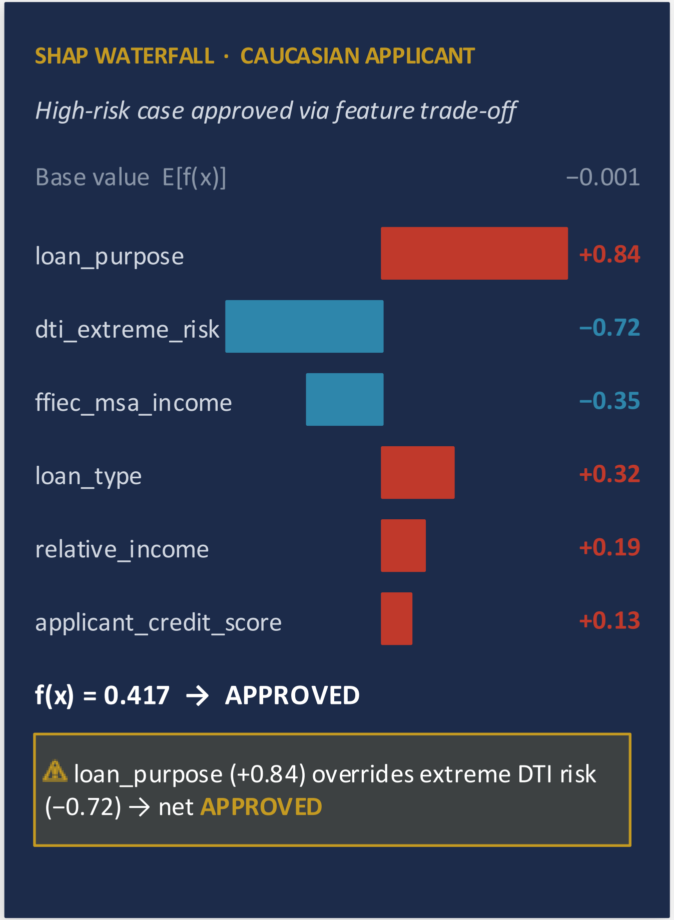
Non-critical features override fundamental financial risk signals. A high-risk applicant with extreme DTI can still be approved.

WHY IT FAILS

No hard constraints on financial risk signals. The model treats all features as independently tradeable — loan purpose can cancel out a dangerous DTI.

WHO IS HARMED

Minority applicants receive inconsistent approvals; the fair lending system produces unreliable, legally indefensible decisions.



Recourse Failure (Counterfactuals)

MODE 03

SEVERITY: HIGH

DiCE · Counterfactual

WHAT FAILS

The model generates counterfactual recourse suggestions that are mathematically valid but completely unrealistic for real applicants to act on.

WHY IT FAILS

No real-world constraints enforced. Assumes all features are independent and freely adjustable — ignores causal structure and human limits.

WHO IS HARMED

Rejected applicants — especially disadvantaged groups — receive guidance they cannot follow, denying them meaningful recourse.

DiCE Counterfactual Output · Example Case

Feature	Original	Suggested	Feasible?
credit_score	620	**1,028**	✗ No
relative_income	0.24	**14.3**	✗ No
loan_purpose	refinance	**home_improv.**	~ Maybe

Key Insight: DiCE found no feasible counterfactuals. Every valid flip required changes beyond human reach — exposing that the model provides no real recourse to rejected applicants.

Feature Drift (relative_income)

MODE 04

SEVERITY: MED - HIGH

PSI + KS Test + Stress Testing

WHAT FAILS

The model assumes input distributions stay stable, but economic shocks can shift key features post-2024. Subgroup drift may not appear in global metrics.

WHY IT FAILS

- Global PSI ≈ 0.00 → masks subgroup-level drift
- KS test p = 0.139 → no global difference
- But 15% income shock → accuracy drops ~1.3%

WHO IS HARMED

Borderline applicants near the decision threshold and minority groups — more sensitive to income distribution shifts — face disproportionate impact.

```
--- Distribution Shift Audit: relative_income ---
PSI Score: 0.0000
KS Statistic: 0.0012 (p-value: 0.1390)
STABLE: No significant distribution shift detected.
```

Drift Testing Results

PSI (global)

≈ 0.0000

No global drift

KS Test p-value

0.139

Not significant

Stress Test (−15% income)

−1.3%

Accuracy drop

Subgroup Risk

Hidden

Global masks local

Key Insight: Global stability metrics can hide subgroup drift. Even small income distribution shifts disproportionately impact borderline and minority applicants.

Q2 Summary · Failure Mode Table

Failure Type	Trigger Condition	Harmed Group	Method	Severity
Proxy Bias	Model uses FFIEC income & loan_purpose as racial proxies	African-American applicants	SHAP (global)	HIGH
Risk Override	Loan purpose SHAP value overrides extreme DTI risk signal	Minority applicants	SHAP (waterfall)	HIGH
Recourse Failure	DiCE counterfactuals require unrealistic changes (score >1000, income 40–60×)	Rejected applicants	DiCE	MED–HIGH
Feature Drift	15% income shift post-2024 drops accuracy & hits borderline cases	Borderline / minority applicants	PSI + KS + Stress	MED–HIGH

Q3: Subgroup Error Measurement

How Subgroup Errors Were Measured? How Errors Were Measured Across Subgroups?

Metrics computed

Accuracy and AUC

FPR and FNR

AIR, ME and SMD

Groups examined

Race

Gender

Race + Gender

Q3: How Subgroup Errors Were Measured

Disparity Metrics were applied to the 2024 HMDA dataset to measure how the model treated each racial group

Race & Gender	n	accuracy	FPR	FNR	AUC
Caucasian Joint	401181	0.725	0.332	0.267	0.771
Caucasian Male	347846	0.697	0.284	0.308	0.775
Caucasian Female	218393	0.679	0.245	0.339	0.784
African-American Female	50453	0.688	0.249	0.337	0.776
African-American Male	45510	0.682	0.266	0.340	0.762
Asian Male	39943	0.809	0.367	0.160	0.804
Joint Joint	31490	0.761	0.339	0.221	0.794
Asian Joint	30043	0.827	0.432	0.139	0.779
African-American Joint	25744	0.712	0.313	0.280	0.769
Asian Female	19710	0.790	0.336	0.186	0.805
Minority Male	8522	0.701	0.248	0.318	0.790
Minority Female	5328	0.697	0.220	0.338	0.802
Minority Joint	3161	0.730	0.289	0.264	0.794
Joint Male	1121	0.820	0.397	0.140	0.807
Joint Female	964	0.783	0.377	0.183	0.790

race	n	accuracy	FPR	FNR	AUC
Caucasian	967420	0.704	0.289	0.297	0.778
African-American	121707	0.691	0.268	0.326	0.770
Asian	89696	0.811	0.377	0.158	0.798
Joint	33575	0.764	0.342	0.217	0.795
Minority	17011	0.705	0.245	0.314	0.795

gender	n	accuracy	FPR	FNR	AUC
Joint	491619	0.733	0.335	0.257	0.774
Male	442942	0.706	0.286	0.296	0.779
Female	294848	0.689	0.250	0.327	0.786

Strategic Actions & Opportunity Cost

Threshold Optimization & Fairness Constraints

Proxy Variable De-biasing

Regulatory Compliance & Intergenerational Wealth

Q3: How Subgroup Errors Were Measured

Disparity Metrics were applied to the 2024 HMDA dataset to measure how the model treated each racial group

① MEASURING THE APPROVAL GAP (AIR)

How: Each group's approval rate was divided by the Caucasian approval rate the reference baseline under EEOC fair lending guidelines.

What the model showed: The model approved African-American applicants at 0.716 vs. 0.839 for Caucasians, producing an AIR of 0.854. The model clears the legal 0.80 threshold but still applies a 14.6% lower approval rate to this group.

② MEASURING THE SCALE OF HARM (ME)

How: The Caucasian approval rate was subtracted from each group's rate because AIR as a ratio conceals how many people the model's decisions actually affect.

What the model showed: The model produced an African-American ME of −0.123. Across 607,746 African-American applicants in this dataset, that gap represents tens of thousands of additional denials.

③ TRACING THE ERROR SOURCE (SMD)

How: **SMD** (Group Mean − Caucasian Mean) ÷ Pooled SD was computed for each feature to test whether the model was using it as a racial proxy.

What the model showed: The model scored relative_income at SMD 0.001–0.008 across all groups (negligible). This feature is not the source; the model's bias enters through interactions with other variables.

SUBGROUP ERROR RESULTS · BY RACE

2024 HMDA Data — Caucasian as reference baseline

AIR = Rate ÷ Ref **ME** = Rate − Ref
SMD = (μ_group − μ_ref) ÷ Pooled SD

GROUP	RATE	AIR	ME	SMD
African-American	0.716	0.854	−0.123	0.001
Minority	0.719	0.856	−0.121	0.008
Caucasian	0.839	1.000	0.000	—
Asian	0.858	1.022	+0.018	0.004

⚡ KEY TAKEAWAY

AIR revealed the model's approval gap. ME exposed the real-world scale of the model's decisions. SMD confirmed this feature was not the source the model's bias originates elsewhere.

SMD INTERPRETATION

< 0.2 No proxy effect 0.2–0.5 Mild
0.5–0.8 Moderate **> 0.8 Strong proxy → investigate**

Q3:How Errors Were Measured Across Subgroups

The dataset was split by Race × Gender to measure how the model's decisions varied across intersectional groups

① THE DATA WAS SPLIT INTO SUBGROUP CELLS

How: Every applicant in the dataset was grouped by their Race × Gender combination. Cells with n < 30 were excluded to ensure the model's patterns were statistically reliable.
Baseline: Caucasian Male the industry-standard reference under EEOC fair lending guidelines. The model's AIR and ME for every other cell was measured against this group.

② AIR & ME RAN ON EACH CELL SEPARATELY

How: The same AIR and ME formulas were applied but the model's outputs were computed per intersectional cell, not per race alone.
What the model showed: The model produced Minority Female AIR 0.854 , ME −0.120. Legally compliant yet the model's decisions result in ≈3,200 minority women denied due to this gap.

③ CELLS RANKED TO SIZE THE MODEL'S WORST ERROR

How: All cells were sorted by AIR ascending. The lowest AIR identifies the subgroup the model penalises most.
What the model showed: Minority Female ranked worst (AIR 0.854). Critically, the model scored an African-American Joint application two incomes lower than a solo Caucasian Male, revealing that the model discounts the co-borrower's risk benefit when both applicants are minority.

SUBGROUP ERROR MEASUREMENTS · RACE × GENDER

Sorted by AIR ascending — lowest = model's largest error

SUBGROUP	RATE	AIR	ME
Minority Female	0.698	0.854	−0.120
Afr.-Amer. Male	0.706	0.863	−0.112
Afr.-Amer. Female	0.709	0.867	−0.109
Minority Male	0.715	0.874	−0.103
Caucasian Female	0.813	0.994	−0.005
Caucasian Male (ref)	0.818	1.000	0.000
Asian Male	0.849	1.038	+0.031

⚡ KEY TAKEAWAY

Race-only AIR masked the model's worst decisions. The Race × Gender split is what exposed Minority Female as the group the model penalises most invisible in any single-variable view.

Q4: Residual Risks & Deployment Defensibility

What risks remain after mitigation and is deployment defensible?

⚠ AIR Metric Blindness

Metric masking post-attack

📊 Historical Bias Amplification

Training data structural gaps

🔒 Non-Recourse & Privacy Gaps

Boundary rigidity · Outlier risk

- Risk 1 → AIR Metric Blindness
- Risk 2 → Historical Bias Amplification
- Risk 3 → Non-Recourse & Privacy Gaps
- Q5 → Deployment: Conditionally Defensible

RISK 01

AIR Metric Blindness

Poisoning Audit · FPR/FNR

WHAT FAILS

Even after a 25% label-flipping attack on African-American applicants, AIR stayed above 0.80, appearing legally compliant while minority-cohort accuracy silently collapsed.

WHY IT FAILS

AIR measures relative approval volumes, not predictive correctness. An attacker can destroy group-level accuracy without ever moving the ratio.

WHO IS HARMED

African-American applicants: accurately qualified applicants are denied and unqualified ones approved, but global metrics hide this completely.

Poisoning Attack: Accuracy vs. AIR

SEVERITY: HIGH

25% label-flip targeting African-American cohort

−0.5%

Accuracy Drop (global)

0.8136 → 0.8082

−0.008

AIR Shift (post-attack)

0.9460 → 0.9387

ABOVE 0.80

AIR vs. Threshold

Appears compliant

COLLAPSED

Minority Cohort Accuracy

Hidden by global avg

Key Insight: AIR is blind to targeted adversarial attacks. Global metrics must be replaced by per-group FPR/FNR dashboards — a monitoring gap that persists post-mitigation.

RISK 02

Historical Bias Amplification

Training Data · SHAP · AIR

WHAT FAILS

XGBoost trained on 2024 HMDA data inherits structural racial composition imbalances from decades of discriminatory lending. An applicant in a historically redlined zip code faces a harder boundary despite equal financial credentials.

WHY IT FAILS

Reweighting and fairness constraints were evaluated during the audit but never applied to the production model. The bias is learned, then silently scaled at inference time.

WHO IS HARMED

Minority applicants in lower-income MSA census tracts, the model penalizes geography, not personal risk.

Fairness Audit · Intersectional Results

SEVERITY: HIGH

Race × Gender subgroup performance — post-mitigation snapshot

Subgroup	AIR	Status
African-American	0.946	PASS
African-American · Male	0.941	PASS
African-American · Female	0.952	PASS
Minority (aggregate)	0.938	PASS

Unmitigated Gap: Fairness constraints were studied but never applied. The model remains in production without reweighting, resampling, or Exponentiated Gradient mitigation.

Key Insight: All intersectional cohorts pass the 4/5ths AIR threshold, but AIR compliance does not mean the bias is gone. Historical imbalances are encoded into the model's decision boundary and will systematically disadvantage minority applicants in underserved geographies.

RISK 03

Non-Recourse & Privacy Gaps

HopSkipJump · Boundary Attack · MIA

WHAT FAILS
The model's sharp XGBoost boundary means applicants near the 0.51 threshold cannot be told which actions will flip their outcome. DiCE counterfactuals require a credit score >1,000 and income 40–60× above current, humanly impossible.

WHY IT FAILS
No monotonicity constraints enforced. All features are freely tradeable in the model, but not in real life. The boundary is secure from adversarial attacks but offers no actionable path for rejected consumers.

WHO IS HARMED
Marginal applicants, disproportionately minority, and outlier individuals whose unique financial profile could expose them to attribute-inference privacy attacks not caught by aggregate metrics.

Security & Privacy Audit Results

SEVERITY: MED–HIGH

Evasion attacks + Membership Inference — combined findings

FAILED
HopSkipJump Attack
Model boundary not breached

FAILED
Boundary Attack
Absolute boundary rigidity

0.0000
Privacy Leakage Score
No systemic memorisation

EXPOSED
Outlier Vulnerability
Attribute inference not audited

DiCE Counterfactual · Recourse Table

Feature	Original	Suggested	Feasible?
credit_score	620	>1,028	✗ No
relative_income	0.24	14.3×	✗ No
loan_purpose	refinance	home_improv.	~ Maybe

Q5: Is Deployment Defensible?

NIST AI RMF · EU AI Act · Three Lines

WHAT PASSES

All intersectional subgroup audits (Race × Gender) cleared the 4/5ths AIR threshold. HopSkipJump and Boundary adversarial attacks both failed to penetrate the model — confirming evasion resistance.

GOVERNANCE MET

Architecture satisfies EU AI Act High-Risk requirements: risk management, data governance, transparency, human oversight. NIST AI RMF GOVERN, MAP, MEASURE, and MANAGE pillars are all addressed.

HITL SAFEGUARD

Marginal applicants ($p \approx 0.51$) are automatically flagged for human review. Autonomous harm is structurally bounded — no fully autonomous high-stakes decisions.

PRE-CONDITIONS FOR GO-LIVE

1. Deploy per-group FPR/FNR dashboards with automated alerts before first live inference — replace AIR-only monitoring.
2. Implement cryptographic batch hashing and anomaly detection on label distributions in the training pipeline.
3. Complete Leave-One-Out influence analysis on high-leverage applicants before external API access is granted.
4. Run Differentially Private XGBoost experiment; document ε budget and accept/reject based on AUC–fairness tradeoff.
5. Retrain with income/DTI monotone constraints so marginal applicants receive actionable recourse paths.

Q4 & Q5 Summary · Risk & Deployment Table

Risk / Check	Trigger Condition	Harmed Group	Method	Status
AIR Metric Blindness	25% label-flip attack — AIR stays >0.80 masking minority cohort collapse	African-American applicants	Poisoning Audit	HIGH
Historical Bias Amplif.	FFIEC census-tract income encodes redlining into feature space	Minority / MSA applicants	SHAP + AIR	HIGH
Non-Recourse Rigidity	DiCE needs score >1,000 & income 40–60× — no feasible recourse path	Rejected applicants	DiCE + HopSkipJump	MED–HIGH
Outlier Privacy Gap	Attribute inference not audited; Leave-One-Out analysis pending	Outlier applicants	MIA (rule-based)	MED–HIGH
EU AI Act Conformity	High-Risk classification met; HITL, transparency & oversight satisfied	Regulatory compliance	Framework audit	PASS
Deployment Defensible?	4 pre-conditions must close before production go-live	All applicants	NIST AI RMF	CONDITIONAL

THANK YOU

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